

SQL Indexing & Triggers - Questions with Answers

Question 1

SQL Indexing & Triggers – Scenario-Task

Scenario 1 – Employee Search Performance

Scenario

The HR department frequently searches employees using their Email Address. As the Employee table has grown to over 10 lakh records, searching has become slow.

Create an appropriate Index to improve the search performance.

Table Required

Employee

Column Name

Data Type

Employee_ID

INT

Employee_Name VARCHAR(100)

Email

VARCHAR(100)

Department

VARCHAR(50)

Salary

DECIMAL(10,2)

Task

-

Create an index on the Email column.

-

Verify that the index is created.

-

Search for an employee using the Email column.

Scenario 2 – Product Search Optimization

Scenario

An e-commerce company frequently searches products based on both Category and Price.

The Product table contains more than 5 million records, and the search queries are taking too much time.

Answer (MySQL)

```
CREATE INDEX idx_employee_email  
ON Employee(Email);
```

```
SHOW INDEX FROM Employee;
```

```
SELECT *  
FROM Employee  
WHERE Email='employee@example.com';
```

Question 2

Create a suitable index to improve the performance.

Table Required

Product

Column Name

Data Type

Product_ID

INT

Product_Name VARCHAR(100)

Category

VARCHAR(50)

Price

DECIMAL(10,2)

Brand

VARCHAR(50)

Task

- Create a Composite Index on Category and Price.

- Display all products in the Electronics category whose price is greater than 50000.

- Explain why a Composite Index is suitable for this query.

Scenario 3 – Employee Audit Log

Scenario

Whenever a new employee joins the company, the HR team wants to automatically maintain an Employee Audit Log.

Every new employee inserted into the Employee table should automatically create a record in the Employee_Log table.

Tables Required

Employee

Column Name

Data Type

Employee_ID

INT

Employee_Name VARCHAR(100)

Department

VARCHAR(50)

Salary

DECIMAL(10,2)

Employee_Log

Answer (MySQL)

```
CREATE INDEX idx_product_category_price
ON Product(Category, Price);
```

```
SELECT *
FROM Product
WHERE Category='Electronics'
AND Price > 50000;
```

```
/* Composite index improves filtering by Category and Price. */
```

Question 3

Column Name

Data Type

Log_ID

INT

Employee_ID

INT

Action_Performed VARCHAR(50)

Action_Time

DATETIME

Task

-

Create an AFTER INSERT Trigger.

-

Whenever a new employee is inserted, automatically insert:

- o Employee_ID

- o Action = 'Inserted'

- o Current Date & Time into Employee_Log.

-

Insert one employee and display both tables.

Scenario 4 – Salary Validation

Scenario

The Finance department has introduced a new rule.

An employee's salary should never become negative.

If any user tries to update a salary with a negative value, the database should reject the update and display an error message.

Table Required

Employee

Column Name

Data Type

Employee_ID

INT

Employee_Name VARCHAR(100)

Salary

DECIMAL(10,2)

Task

-

Create a BEFORE UPDATE Trigger.

-

Prevent updating a salary if it is less than 0.

-

Display the message:

Answer (MySQL)

```
DELIMITER //
CREATE TRIGGER trg_employee_after_insert
AFTER INSERT ON Employee
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Log
    (
        Employee_ID,
        Action_Performed,
        Action_Time
    )
    VALUES
    (
        NEW.Employee_ID,
```

```
        'Inserted',
        NOW()
    );
END//
DELIMITER ;

INSERT INTO Employee
VALUES (101, 'Rahul', 'HR', 45000);

SELECT * FROM Employee;
SELECT * FROM Employee_Log;
```

Question 4

Salary cannot be negative

-

Test the trigger using an UPDATE statement.

Scenario 5 – Customer Email Validation

Scenario

A banking application stores customer details.

Business requirements:

1. No two customers should have the same Email Address.
2. The customer's Age must be at least 18 years before registration.

Implement the required database objects.

Table Required

Customer

Column Name

Data Type

Customer_ID

INT

Customer_Name VARCHAR(100)

Email

VARCHAR(100)

Age

INT

Task

-

Create a Unique Index on the Email column.

-

Create a BEFORE INSERT Trigger.

-

Reject the insert if Age is less than 18.

-

Display the message:

Customer must be at least 18 years old

-

Test the trigger by inserting both valid and invalid records.

Answer (MySQL)

```
DELIMITER //
CREATE TRIGGER trg_salary_validation
BEFORE UPDATE ON Employee
FOR EACH ROW
BEGIN
    IF NEW.Salary < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='Salary cannot be negative';
    END IF;
END//
DELIMITER ;

UPDATE Employee
SET Salary=-5000
WHERE Employee_ID=101;
```